
Algorithmen und Wahrscheinlichkeit

Exercise Session 4

Exercise 1 – *Basic probability*

Describe the probability space (including the probability function) for each of the following tasks.

- (a) We roll a red and a black die (both dice are standard 6-sided dice) and define the following events A, B, C, D

- A = 'The red die shows an even number.'
- B = 'The black die shows a 5.'
- C = 'The product of the numbers is a prime number.'
- D = 'The sum of the numbers is divisible by 3.'

Here, the notation A = 'statement' means that A is the set of all elementary events for which the statement is true. Calculate the probabilities $\Pr[X]$ for all $X \in \{A, B, C, D\}$. Calculate $\Pr[A|C]$. Are the events A, B (mathematically!) independent? What is the behavior for events B, C , and C, D ? (So you have to determine whether each pair is independent for a total of three pairs of events.)

- (b) You and your two friends have three vegan hot dogs. One of them is completely intact, one has a bite on (exactly) one side, and the third has a bite on both sides. To divide the hot dogs fairly, you are randomly assigned one of the three hot dogs. You see a random end of your hot dog and are happy that it hasn't been bitten. What is the probability that you are lucky and have received the completely intact hot dog? Note: Don't forget to describe the probability space. What assumptions, implicit in the text, are you making?
- (c) David loves flipping coins. Whenever he sees the sequence tails, heads, heads, tails (in direct succession), he is particularly happy. Suppose David flips his coin 10 times. What is the probability that his favorite sequence occurs in the 10 flips? More precisely, we denote by $M_1, \dots, M_{10}$ the results of the 10 coin flips and want to calculate the probability that there exists an $i \in \{1, 2, \dots, 7\}$ such that

$$M_i = H, M_{i+1} = T, M_{i+2} = T, M_{i+3} = H. \quad (1)$$

Hint: For $i = 1, \dots, 7$, let E_i be the event that (1) occurs for i . What is $\Pr[E_1 \cup E_2 \cup \dots \cup E_7]$? Use the inclusion-exclusion principle.